Sentiment Analysis in Social Media Data for Depression Detection Using Deep Learning

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DEDICATION

I dedicate this work to those who silently endure the challenges of depression. Your resilience inspires every step of this research, and may this study contribute to a better understanding and more compassionate support.

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First and foremost, I am deeply grateful to Allah Almighty for His countless blessings, guidance, and strength throughout this journey.

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LIST OF SYMBOLS, ABBREVIATIONS AND ACRONYMS

AB Adaptive Boosting. 13

AP Affinity Propagation. 13

AUC Area Under the Curve. 33

BiGRU Bidirectional Gated Recurrent Unit. 17

BiLSTM Bidirectional Long Short-Term Memory. 15, 16

BoSE Bag of Sub Emotions. 13

BoW Bag of Words. 13

BP Bagging Predictors. 13

CNN Convolutional Neural Network. 8, 15, 16, 18

DB Data-based. 14

DL Deep Learning. 1–4, 12, 14, 15

DT Decision Tree. 12, 13

FAN Feature Attention Network. 16

GB Gradient Boosting. 13

GBDT Gradient Boosting Decision Tree. 15

gSC Global Strength Classifier. 14

HAN Hierarchical Attention Network. 16

KB Knowledge Based. 14

kNN K-Nearest Neighbors. 12, 14

LIME Local Interpretable Model-agnostic Explanations. 8

LIWC Linguistic Inquiry and Word Count. 7, 8, 14

LR Logistic Regression. 8, 12, 13

LXMERT Learning Cross-Modality Encoder Representations from Transformers. 25

ML Machine Learning. 3, 12–16

MLP Multilayer Perceptrons. 12, 13

NB Naïve Bayes. 8

NLP Natural Language Processing. 2, 8

OCC One-Class Classification. 14

OCC-kSS One-Class Classification k-Strongest Strengths. 14

POS Part of Speech. 14

RF Random Forest. 8, 13

RNN Recurrent Neural Network. 3, 15

SHAP SHapley Additive exPlanation. 8

SVM Support Vector Machine. 8, 12–15

TF-IDF Term Frequency-Inverse Document Frequency. 8, 14, 15

ABSTRACT

Depression is a widespread mental health challenge affecting millions globally, with serious social and economic consequences. With the rise of social media, these platforms have become useful tools for observing mental health patterns through user-generated content. However, many current deep learning approaches for depression detection focus solely on text, neglecting the potential cues present in images. Furthermore, while some multimodal techniques exist, they often fail to effectively integrate the contextual relationships between text and images, limiting their ability to capture comprehensive depressive indicators. To fill this research gap, a custom deep learning architecture is developed in this work that utilizes learnable context vectors to capture contextual relationships between text and image data. The proposed model utilizes the CLIP model with learnable context vectors to encode both modalities, allowing dynamic adaptation to linguistic and visual patterns related to depression. A cross-modal encoder fuses the embeddings followed by processing through a transformer-based architecture. The model is evaluated on a publicly available multimodal Twitter dataset. The results show that our model exceeds the performance of existing state-of-the-art techniques, achieving an AUC of 0.9922, precision of 0.9765, recall of 0.9479, and an F1-score of 0.9619 across 5-fold cross-validation. Additionally, the integration of learnable context vectors improves accuracy and robustness. This research underscores how multimodal deep learning can enhance mental health monitoring and support early intervention, stressing the value of context-aware modeling in managing the complexities of real-world social media content.

Keywords: CLIP framework, context vectors, deep learning, depression detection, mental health, multimodal fusion, social media, transformer network.

CHAPTER 1: INTRODUCTION AND MOTIVATION

Affecting millions globally, depression is a widespread mental illness that has serious emotional, social, and financial impacts. Social media's emergence in recent years has revolutionized communication by giving people unmatched access to real-time insights into human emotions and behavior. New methods for monitoring mental health, especially for identifying signs of depression at an early stage, have been made possible by the abundance of user-generated content.

Pattern recognition and data analysis in a variety of fields have been transformed by the quick development of Deep Learning (DL). DL algorithms can learn complex representations from unstructured text, photos, and metadata uploaded on social media networks, which is useful for detecting depression. While text-based analysis is the main emphasis of the majority of current methodologies, social media posts frequently contain multimodal content, such as text and images, which when combined, offer a deeper context for comprehending user emotions. Due to the differing structures of text and image modalities and the need for multiple feature extraction methods, efficiently integrating and evaluating multimodal data is still difficult.

In addition to technical factors, DL in this area faces challenges related to social media data. These challenges include combining different types of data, managing noise, and dealing with language diversity and varying contexts. Social media posts often include sarcasm or ambiguous wording, which can confuse standard sentiment analysis methods. Addressing these issues is essential to achieving more accurate and reliable model performance.

The modern, interconnected world has seen a noticeable rise in mental health awareness. Automated depression detection systems go beyond just technical progress. They offer important benefits by helping with early diagnosis, reducing stigma, and supporting personalized interventions. As we develop DL methods for mental health, it is crucial to ensure fairness, privacy, and ethical data use. Advanced computational techniques have the potential to support more proactive and preventive strategies in mental health care.

1.1 Motivation

This research is motivated by the pressing challenge of global mental health, with depression ranking among the most serious disorders. Social media platforms, where individuals regularly express their emotions and thoughts, offer a valuable data source for identifying early signs of mental issues. Although DL models show great potential in text and sentiment analysis, detecting depression in the informal and diverse language of social media is still challenging. Additionally, the limited use of multimodal data reduces the effectiveness of current methods.

These challenges, along with the benefits of early intervention, highlight the need for reliable, real-time depression detection systems that use both text and image data.

1.2 Problem Statement and Challenges

Problem Statement: While current depression detection models leveraging multimodal social media data have made progress, they struggle with subtle or ambiguous posts because they fail to effectively integrate text and image context, limiting their accuracy and reliability. This thesis aims to develop a DL framework that combines text, images, and contextual information to improve both accuracy and robustness.

Researchers have identified and addressed the following key challenges to enhance the performance of depression detection models.

1. *Ambiguity in Language and Expression*
   - **Challenge:** Social media language is usually informal and often includes slang, abbreviations, sarcasm, and cultural references. This makes it challenging to accurately interpret sentiment and identify mental health indicators.
   - **Potential Solution:** Leverage modern Natural Language Processing (NLP) methods and contextual embeddings (e.g., BERT, RoBERTa) to capture subtle linguistic patterns and context-dependent meanings.
2. *Data Imbalance and Noise*
   - **Challenge:** Depression-related content may be underrepresented compared to general social media chatter, and posts are often noisy and unstructured.
   - **Potential Solution:** Implement data augmentation and sampling techniques, along with effective preprocessing methods, to better manage imbalanced and noisy datasets.
3. *Multimodal Data Integration*

- **Challenge:** Combining multimodal data, such as text and images, is challenging because each type has its own structure and features.
- **Potential Solution:** Use advanced fusion techniques, like attention mechanisms and cross-modal learning, to merge textual and visual information for enhanced detection performance.

4. *Temporal Dynamics of Depression*

   - **Challenge:** Mental health disorders like depression are dynamic in nature, and the way people express symptoms online also evolves. Capturing these changes is important for accurate detection.
   - **Potential Solution:** Utilize time-aware models like Recurrent Neural Networks (RNNs) or transformer networks with temporal attention, which can help track evolving symptoms and context shifts.

1.3 Objectives and Contributions

1.3.1 Objectives

Our research to detect depression from social media content using DL is driven by the following objectives:

- **Literature and Dataset Review:** Perform a thorough review of depression detection on social media. This review will examine how the field has evolved from traditional Machine Learning (ML) methods to modern DL approaches. It will cover common architectures, datasets, ethical issues, major applications, and current research challenges.
- **Model Development:** Develop a DL model that uses both text and image data. The model will incorporate learnable context vectors to improve its understanding of context.
- **Evaluation and Benchmarking:** Evaluate the model using standard performance metrics such as accuracy, precision, recall, F1 score, AUC, and the confusion matrix, and compare the results with those of existing methods.

1.3.2 Contributions

1. **Contribution 1:** *Depression Detection in Social Media: A Comprehensive Review of Machine Learning and Deep Learning Techniques* – This work summarizes key advancements in the field and addresses gaps in earlier surveys by providing

a detailed analysis of datasets, application areas, and a general framework for depression detection.

Previous reviews have often focused only on selected studies, datasets, or methods, and they typically lack a thorough discussion of application areas, general architectures, and the evolution of depression detection techniques. In this study, we review important research, classify datasets, and explore how methodologies have progressed. Additionally, we propose a general framework for depression detection and discuss emerging challenges in the field.

2. **Contribution 2:** *Context-Aware Multimodal Framework* – Development of a novel DL framework that integrates text and image features while leveraging learnable context vectors to enhance contextual understanding.

1.4 Thesis Organization

This thesis is structured as follows, each addressing key aspects of depression detection in social media using DL.

1.4.1 Chapter 2: Background and Theoretical Foundations

Chapter 2 reviews the theoretical foundations of depression detection in social media. It covers key methods ranging from lexicon-based models to advanced deep learning, while also addressing ethical concerns and open research areas.

1.4.2 Chapter 3: Literature Review

Chapter 3 provides an overview of ML and DL techniques for depression detection in social media. We discuss key models, feature extraction methods, and evaluation frameworks while highlighting research gaps.

1.4.3 Chapter 4: Design and Methodology

Chapter 4 presents the design and methodology of the proposed deep learning framework for detecting depression using multimodal social media content. It outlines the overall architecture of the model, which integrates text, images, and their context from social media posts.

1.4.4 Chapter 5: Experimental Setup

Chapter 5 provides a detailed description of the experimental setup for our model. It includes information about the dataset utilized for training and evaluation, the computational environment for GPU-based training, implementation details, and the hyperparameter configurations. Additionally, it also describes the metrics employed to evaluate the model's performance.

1.4.5 Chapter 6: Results and Analysis

Chapter 6 discusses the evaluation results of the proposed model, offering in-depth analysis using metrics such as accuracy, precision, recall, F1 score, and other relevant metrics. Additionally, results are compared against state-of-the-art models to highlight improvements.

1.4.6 Chapter 7: Conclusion and Future Work

The final chapter, Chapter 7, provides an overview of the main contributions of this research, highlighting the impact of multimodal deep learning in depression detection. It discusses the effectiveness of learnable context vectors in enhancing model performance and outlines potential areas for future exploration.

CHAPTER 2: THEORETICAL FOUNDATIONS AND EXISTING APPROACHES IN DEPRESSION DETECTION IN SOCIAL MEDIA

This chapter explores the theoretical basis for detecting depression through social media and examines current methodologies. It outlines the primary techniques used for text and multi-modal analysis, reviews benchmark datasets, and discusses the remaining research areas that need to be addressed to build more accurate and ethical depression detection models.

2.1 Theoretical Foundations

2.1.1 Understanding Depression

Depression is a psychological condition marked by ongoing sadness, reduced interest in daily activities, and difficulties with thinking or concentration. Clinically, it is diagnosed based on the Diagnostic and Statistical Manual of Mental Disorders (DSM-5) criteria, which include symptoms such as fatigue, insomnia, changes in appetite, and suicidal thoughts [1]. Clinical diagnosis requires mental health professionals to conduct detailed interviews and assessments. Automated depression detection, on the other hand, utilizes computational analysis of digital data to detect indicators of depressive symptoms. While this approach mainly focuses on text, it is increasingly incorporating multimodal content. It is critical to recognize that automated detection functions as a screening or risk assessment tool, whereas clinical diagnosis requires evaluation by a qualified professional.

2.1.2 Social Media's Impact on Mental Health Monitoring

Social media platforms like Reddit, Twitter, Instagram, and Facebook offer meaningful perspectives into how users express emotions and mental states in digital spaces. Users frequently share their daily experiences, thoughts, and feelings, generating a vast repository of textual and visual data that reflects their psychological state.

Research shows that people experiencing depression often display distinctive online behaviors and linguistic patterns. These may include reduced social interaction, such as fewer replies and increased withdrawal from online engagement. Linguistically, their language tends to exhibit negative sentiment, pessimism [2], and markers of rumination [3], along with a higher frequency of personal pronouns like "I" [4].

To analyze these patterns, researchers implement multiple computational techniques, including n-grams, topic modeling, and sentiment analysis. Researchers have extensively employed these approaches in analyzing text-based social media content to infer mental health states and identify potential indicators of depression.

2.1.3 Ethical and Privacy Considerations in Depression Detection

Research on depression detection using social media content can provide useful insights, but it also raises important ethical concerns like privacy, consent, and potential misuse. Many users do not realize that their public posts are being used for research, which raises concerns about informed consent and data privacy. To address these problems, researchers should anonymize the data, get ethical approvals, and, when possible, obtain user consent. Additionally, inaccurate or biased predictions might lead to stigmatization or exploitation by employers or insurance companies. Therefore, it is important to use strong anonymization methods, build transparent models, and work closely with ethicists and mental health professionals. Although social media-based mental health predictions could improve early detection and intervention, researchers must balance these benefits with ethical responsibilities and prioritize user rights and well-being.

2.2 Existing Approaches in Depression Detection

2.2.1 Traditional (Lexicon and Rule-Based) Approaches

The majority of early approaches to sentiment analysis and depression identification were lexicon-based and rule-based. For example, sentiment analysis with tools like Linguistic Inquiry and Word Count (LIWC), VADER, and the NRC Emotion Lexicon counts words associated with specific emotions in a text. Rule-based systems, on the other hand, classify depression using sentiment scores that are assigned based on predefined language rules [5]. Despite their ease of use and computational efficiency, these conventional approaches have several drawbacks. They often struggle to understand nuances like sarcasm, irony, or contextual variations in word meaning.

2.2.2 Machine Learning-Based Approaches

Machine learning has led to more advanced techniques for depression detection. These methods start with feature extraction and then use supervised learning. For instance, Term Frequency-Inverse Document Frequency (TF-IDF) [6] measures word importance. LIWC [7] is another common tool for feature extraction. Models like Naïve Bayes (NB), Logistic Regression (LR), Random Forest (RF), and Support Vector Machines (SVMs) [6] are then applied. These approaches learn patterns from labeled data and capture relationships in text. However, they require extensive feature engineering. They also depend on high-quality labeled data, which is often hard and expensive to obtain. Traditional machine learning models may struggle with long-range dependencies and subtle context in text.

2.2.3 Deep Learning-Based Approaches

Deep learning has transformed many fields, including NLP. It is now widely used in depression detection research. Neural architectures like Convolutional Neural Networks (CNNs) [8] and Long Short-Term Memory networks (LSTMs) [9] have shown good results in this area. Recently, Transformer models like BERT have reached state-of-the-art performance [10]. These models use attention mechanisms to capture context in text. They can also be pre-trained on large text collections, which helps with transfer learning.

2.2.4 Multimodal Approaches and Explainable AI

Recent research has integrated multimodal techniques in mental health analysis. This means combining text data with images or user behavior. For example, one study analyzed Instagram posts by integrating captions and visual content to detect signs of depression [11]. Multimodal analysis uses different data types to improve detection accuracy. Recent work also emphasizes transparency by making depression detection models more interpretable. Techniques like attention mechanisms [12], SHapley Additive exPlanation (SHAP), and Local Interpretable Model-agnostic Explanations (LIME) [13] help explain how these models make decisions.

2.3 Datasets

The effectiveness of depression detection models depends on the availability of high-quality, annotated datasets. Table 2.1 summarizes key benchmark datasets frequently utilized in studies on depression detection.

Table 2.1. Overview of Commonly Used Social Media Datasets for Depression Detection

Dataset	**Platform**	**Data Type**	**Description**
CLPsych 2015 [14]	Twitter	Text	Includes 327 depressed users and 572 control users.
RSDD [15]	Reddit	Text	Contains 9,210 users diagnosed with depression and 107,274 control users.
SMHD [16]	Reddit	Text	Contains 14,139 users identified as having depression and 335,952 control users.
Gui et al. [17]	Twitter	Text & Images	Multimodal dataset containing 1,402 depressed users and 5,160 control users.
multiRedditDep [18]	Reddit	Text & Images	Extensive multimodal dataset that includes 6.6 million posts from depressed users and 8.1 million posts from non-depressed users.

2.4 Open Research Areas

As research on depression detection through social media continues to evolve, several promising directions emerge. Addressing these areas can contribute to more reliable, inclusive, and effective mental health monitoring systems.

2.4.1 Enhancing Dataset Representativeness and Generalization

Existing models have limited generalizability because current datasets often over-represent populations from the USA and Canada [19]. Future work should focus on creating more representative datasets covering a broader spectrum of age demographics, ethnicities, cultural backgrounds, and geographic locations. This diversity is essential for building models that work reliably across different populations and can be used on a global scale.

2.4.2 Multimodal Strategies

While many studies still focus on text analysis, researchers are increasingly using multimodal methods. These methods combine visual, auditory, and other types of data. This emerging approach could improve detection accuracy by taking advantage of different data strengths. Future research may explore advanced fusion techniques and new model designs to capture the complex patterns of depression.

2.4.3 Expanding Multilingual Capabilities

Current depression detection models are often designed for specific languages [20], which limits their global use. Future research should focus on building robust multilingual models that can detect depression across different languages and cultures. This will require better cross-lingual transfer learning and the integration of culturally relevant mental health indicators to improve accessibility and effectiveness.

2.4.4 Expanding to Underexplored Social Media Platforms

Many studies focus on Twitter and Reddit because they provide accessible and structured text data. However, platforms like Facebook, TikTok, and Instagram offer rich multimedia content. This content can provide valuable insights into depression. Future research should explore ways to extract and analyze these signals. It is important to consider each platform's unique content and user interactions.

2.4.5 Building Unified Depression Detection Models

Developing one model that detects depression across different social media platforms is still a challenge. Future research should explore domain adaptation techniques and platform-specific feature learning. This approach could improve both consistency and scalability. Such models may offer a general way to identify mental health conditions across various online environments.

2.4.6 Integrating Cross-Platform User Behavior Analysis

People often have several social media accounts and behave differently on each platform. Future research should develop methods to combine data from all these sources. Using multiple data sources can improve depression detection accuracy and provide a clearer picture of mental health trends.

2.5 Summary

This chapter examines the theories and methods used in depression detection in social media. It starts by discussing the psychological basis of depression and the role of social media in mental health research. The chapter reviews key computational methods, including lexicon-based approaches, machine learning models, and deep learning techniques like transformers. It also looks at the use of multimodal data and explainable AI. In addition, it presents benchmark datasets commonly used in this field and highlights research challenges such as dataset diversity and multilingual support. The chapter concludes by suggesting future directions to improve accuracy and generalizability in automated depression detection systems.

CHAPTER 3: LITERATURE REVIEW

A detailed analysis of previous studies on depression detection via social media using ML and DL techniques is presented in this chapter. While the previous chapter introduced the theoretical underpinnings and key concepts of depression detection, this section critically examines recent methodologies, performance metrics, and benchmarking strategies in the field. By comparing different approaches, we identify the limitations of current models and highlight the research gap that our study aims to address.

3.1 Machine Learning Approaches

This section explores the use of conventional ML approaches to identify depression in social media data. Frequently used before deep learning became widely used, these techniques established the fundamentals of feature engineering and classification techniques for this challenging task. We look at several important papers that highlight the variety of ML techniques for depression detection, with an emphasis on feature extraction techniques, classification algorithms, and evaluation frameworks.

3.1.1 Feature Engineering and Linguistic Analysis

Early ML methods relied on careful feature engineering. These methods often use language analysis and sentiment detection. Islam et al. [21] studied Facebook post data. The study applied simple preprocessing steps like tokenization, stemming, and sentiment analysis. The approach combined classifiers such as SVMs, Decision Trees (DTs), and K-Nearest Neighborss (kNNs). The results showed that the DT model worked best, especially for recall and F-measure. Time series analysis revealed that depressive posts were more common in the morning. Seasonal trends were observed in late summer.

Fatima et al. [22] studied postpartum depression detection using Reddit posts. They used the LIWC tool to extract 94 features. Then, they reduced these to 20 key features with LASSO. Their two-layer machine learning model included SVMs, Multilayer Perceptronss (MLPs), and LR. The model showed that linguistic analysis works well. It achieved 91.7% accuracy for general depressive content and 86.91% for

PPD-specific content. LR worked best for general content, while MLP performed best for PPD-specific content.

Katchapakirin et al. [23] studied the Thai Facebook community using linguistic insights. They used NLP tools and sentiment analysis to extract features. They compared three models: SVM, Random Forest, and deep learning algorithms. All models outperformed a simple majority vote baseline. Random Forest performed better than SVM, while the deep learning model achieved the highest accuracy. Their results highlighted that negative sentiment was an important indicator. Posts showing negative sentiment without emoticons and with strict privacy settings were linked to a higher chance of depression. In contrast, active social engagement was linked to a lower chance.

3.1.2 Emotion and Semantic Approaches

Beyond general sentiment, more nuanced approaches have examined emotional dimensions and semantic representations. Aragón et al. [24] used fine-grained emotional data derived from social media posts. They employed a two-step process. First, they used a lexicon of core emotions paired with sentiment labels. Then, they applied Affinity Propagation (AP) clustering on FastText sub-word embeddings to form detailed emotional subgroups. Their Bag of Sub Emotions (BoSE) models—which represent emotion occurrence and sequence histograms—significantly outperformed traditional Bag of Words (BoW) models when classified with a linear SVM. Adding sentiment data further improved performance, emphasizing the importance of capturing specific emotional nuances in depressive communication.

3.1.3 Ensemble Methods and Feature Set Combination

Several studies have highlighted the effectiveness of ensemble methods and the benefits of combining diverse feature sets. Chiong et al. [25], for instance, implemented an ML approach leveraging Twitter posts for depression detection. They explored a comprehensive feature set derived from sentiment lexicons—namely, SentiWordNet and SenticNet—as well as tweet characteristics. The authors compared various classifiers including SVMs, DTs, MLPs, and ensemble methods such as RF, Bagging Predictors (BP), Adaptive Boosting (AB), and Gradient Boosting (GB). They observed that SenticNet-based features outperformed those from SentiWordNet, and further improvements were achieved by integrating content-based features. Among the models, GB emerged as the top-performing, achieving accuracy exceeding 98% across datasets and effectively mitigating class imbalance issues, especially in datasets with skewed distributions.

Islam et al. [21] also employed ensemble methods alongside individual classifiers, further demonstrating their potential in this domain. Similarly, Adarsh et al. [26]

introduced an ensemble of SVM and kNN for suicidal thought detection on Reddit, achieving high performance metrics. The constant success of ensemble techniques implies that they are capable of handling the complex structure of social media content as well as the nuances of depression identification.

3.1.4 Contextual and User-Centric Features

Some studies have expanded feature sets beyond just text to include contextual and user-level information. Ding et al. [27] designed a system to identify signs of depression among college students based on data from Weibo. Their approach integrated features like emoji frequency, word frequency, post count, and network characteristics (followers and followees). They used an integrated SVM with AdaBoost. This model showed better performance compared to a standard SVM and other classifiers. Their work illustrates the significance of considering user-level features and the timing of data collection.

Liaw et al. [28] highlighted the significance of user engagement features in detecting depression on Twitter. They built extensive datasets and used a wide range of features, including textual content, user activity, network interactions, and engagement metrics. Their results indicated that XGBoost outperformed all other evaluated models. Notably, user engagement features, especially the presence of depression keywords in liked tweets, significantly improved model accuracy. This finding emphasizes the rich information contained in user interactions beyond just the text of the posts.

3.1.5 Alternative Classification Approaches

Aguilera et al. [29] explored a novel approach that moves beyond traditional binary classification. They applied One-Class Classification (OCC) techniques such as One-Class Classification k-Strongest Strengths (OCC-kSS) and Global Strength Classifier (gSC) to detect mental illnesses, including depression, in social media posts. Their method used domain-specific term frequencies from Knowledge Based (KB) and Data-based (DB) lexicons. They found that gSC, especially when combined with KB and DB lexicons, outperformed both traditional OCC and binary classifiers. The resilience of gSC with limited training data suggests its usefulness in real-world scenarios with scarce labeled data.

3.1.6 Large-Scale Data and Feature Importance

Skaik et al. [30] leveraged large-scale Twitter datasets to train and evaluate both classical ML and DL models. Their study, employing a wide range of features including statistical measures, TF-IDF, LIWC categories, Part of Speech (POS) tags, topic modeling, and sentiment analysis, found that Gradient Boosting Decision Tree (GBDT) achieved the highest F1-score among ML models, even outperforming some DL approaches in certain metrics. This highlights the continued relevance and effectiveness of well-engineered ML models, particularly when combined with comprehensive feature sets and evaluated on large datasets.

3.2 Deep Learning Approaches

With a foundation in ML, research in social media–driven depression detection is now increasingly leveraging DL methods. DL approaches have the benefit of automatically extracting complex features directly from raw data. These methods can capture intricate patterns in language, user behavior, and even multimodal content that traditional feature engineering might overlook. This section examines several prominent deep learning techniques used for this task by discussing their network architectures, feature utilization, and performance characteristics.

3.2.1 RNNs and CNNs for Sequential and Feature Learning

Fundamental DL architectures like RNNs and CNNs have been effectively used for depression detection. Gamaarachchige et al. [8] introduced a multi-task learning method to detect PTSD and depression using social media data. They employed a multi-channel CNN architecture with optimized vocabulary generation based on TF-IDF. This approach outperformed traditional RNNs and SVMs and demonstrated strong performance across multiple metrics, including accuracy, precision, recall, F1-score, and AUC. Their work demonstrates that CNNs can capture important features from social media text, especially when combined with multi-task learning and improved data preprocessing. The CBA model by Thekkekara et al. [31] also integrates CNNs for feature extraction together with Bidirectional Long Short-Term Memory (BiLSTM) networks.

3.2.2 Multi-Task Learning and Neuro-Symbolic Approaches

Multi-task learning, as seen in Gamaarachchige et al. [8], Wang et al. [32], and Ilias et al. [33], is a promising strategy for enhancing depression detection models. Ilias et al. [33] specifically investigated multi-task learning for joint depression and stress detection, introducing Double Encoders and Attention Fusion Network MTL architectures utilizing BERT tokenizers. Their multi-task learning models outperformed both single-task and transfer learning approaches, highlighting the advantages of joint learning for related mental health conditions.

Dou et al. [34] presented a novel Neuro-Symbolic AI framework called TAM-SenticNet for early depression identification. This framework combines symbolic logic from SenticNet with a Time-Aware Affective Memories (TAM) network. The TAM network includes modules for semantic processing, integration, T-LSTM, and emotion processing. TAM-SenticNet was developed to improve both the interpretability and accuracy of depression detection models. Evaluated on the CLEF eRisk datasets, it outperformed other models, highlighting the promise of neuro-symbolic approaches to connect data-driven deep learning with knowledge-based reasoning in mental health detection.

3.2.3 Attention-Based Architectures

Attention mechanisms have become increasingly common in deep learning models for depression detection, helping the model to concentrate on key parts of the input data. Song et al. [35] introduced a Feature Attention Network (FAN) that uses an attention mechanism at the post level. This method effectively prioritizes relevant posts from user histories when classifying depression using the Reddit Self-reported Depression Diagnosis (RSDD) dataset. FAN combines four feature networks based on psychological theories, which focus on depressive symptoms, sentiment, ruminative thinking, and writing style. This approach demonstrates the value of aligning model architecture with domain-specific knowledge. The model yielded an F1 score of 0.56 during evaluation.

Sekulić et al. [36] adapted the Hierarchical Attention Network (HAN) for depression detection using the SMHD dataset. HAN was initially developed to handle document classification tasks and employs GRU-based sequence encoders along with attention mechanisms. The model treats each user as a sequence of social media posts. It outperformed traditional ML baselines by obtaining an F1-score of 0.682. This performance shows that it can successfully learn and represent relevant linguistic cues, such as affective language and personal pronouns, as supported by attention weight analysis. Similarly, Han et al. [37] used HAN but extended it to incorporate metaphorical concept mappings (MCMs) alongside tweet content, using BERT embeddings for both. Their enhanced HAN model applied attention weights for both tweet and MCM features and achieved an average F1-score of 0.972. This result demonstrates the potential of combining textual content with higher-level semantic representations while emphasizing explainability through attention mechanisms.

Thekkekara et al. [31] further extended attention mechanisms by proposing a CNN-BiLSTM Attention (CBA) model. Their approach combines CNNs for feature extraction with BiLSTMs for sequential modeling and integrates an attention mechanism to focus on crucial linguistic components in social media texts. Evaluation was conducted using the CLEF2017 eRisk dataset and outperformed various baseline models, obtaining an accuracy of 96.71% and an F1-macro score of 0.89. These results demonstrate the power of hybrid CNN-RNN architectures enhanced with attention for precise depression detection.

3.2.4 Multimodal Deep Learning Approaches for Richer User Representations

Recognizing that depression manifests in multifaceted ways, many studies have explored multimodal deep learning approaches that integrate various data modalities beyond text. An et al. [38] proposed the Multimodal Topic-Enriched Auxiliary Learning (MTAL) framework that combines textual and visual inputs. In this method, BERT and VGG models are used to encode text and images, respectively, features are fused through an LSTM, and modality-agnostic topic modeling is employed as an auxiliary task to enhance depression prediction. Their model obtained an accuracy of 84.2% on a balanced dataset. Similarly, Li et al. [28] presented a Multimodal Hierarchical Attention (MHA) model for detecting depression symptoms. MHA enhances attention by modeling interactions both within and across different modalities, including text, images, and auxiliary data. It utilizes ResNet-18 for extracting image features, TextCNN for text, and incorporates additional data like user interactions. Evaluated on Weibo data, MHA outperformed baseline models with an F1-score of 0.927, highlighting the versatility of hierarchical attention in multimodal depression detection.

Wang et al. [32] developed FusionNet, a multitask learning framework that integrates text, social behavior, and image data for depression detection on Weibo. They used XLNet for text encoding and applied statistical methods for processing social behavior and image features. FusionNet employed a Bidirectional Gated Recurrent Unit (BiGRU) with attention for classification. Their multitask strategy, which optimizes both word vector tasks and statistical feature tasks, obtained an F1-score of 0.97 on the newly created WU3D dataset, outperforming traditional machine learning models.

Yadav et al. [39] further explored multimodal aspects by focusing on fine-grained depression symptom detection in memes, a rich source of multimodal communication. They used pre-trained ResNet-152 for visual encoding and BERT for textual encoding. The authors implemented 2D adaptive average pooling and orthogonal regularization to boost feature expressiveness and employed conditional adaptive gating for multimodal fusion. Their model was evaluated on the RESTORE depressive meme dataset and achieved promising results with an F1-score of 0.65. These results highlight the capability of multimodal deep learning for detecting nuanced depression symptoms in meme content.

Bucur et al. [12] explored multimodal depression detection by focusing on sequences of text and images. Their transformer-based method utilized pre-trained CLIP and EmoBERTa models, followed by a cross-modal encoder and a Time2VecTransformer to capture temporal patterns. The Time2VecTransformer obtained an F1-score of 0.931

on a Twitter dataset, demonstrating the benefit of incorporating temporal awareness into multimodal models.

Anshul et al. [40] proposed a Multimodal Feature-based Ensemble Learning (MFEL) approach that integrates textual, visual, and user-specific features. They extracted both intrinsic and extrinsic textual features and used a CNN-based model known as VNN for visual feature extraction. An ensemble technique then combined the outputs of multiple classifiers. Their method obtained an F1-score of 0.91 on the COVID-19 dataset.

Zafar et al. [41] introduced Multi-Explainable TemporalNet (METN), a multimodal temporal network that combines a Temporal Convolutional Network (TCN) with a multimodal transformer framework. They leveraged pre-existing models such as CLIP, EmoBERTa, and ViLBERT for cross-modal alignment. METN also incorporated attention maps for improved interpretability. The model achieved F1-scores of 0.945 on a Twitter dataset and 0.913 on a Reddit dataset, highlighting the significance of temporal modeling and explainability in multimodal approaches.

Table 3.1 summarizes recent studies on depression detection using multimodal approaches, highlighting the datasets used, model architectures, reported F1 scores, and key limitations of each approach.

Paper	**Year**	**Dataset**	**Model**	**F1 Score**	**Limitations**
[32]	2022	Weibo	Bi-GRU with attention	0.977	Uses low-level image features.
[28]	2023	Weibo	MHA	0.927	Model fails if depressive signs are vague or require contextual understanding.
[39]	2023	Twitter, Reddit	ResNet-152 and BERT	0.651	Ignores accompanying post text and temporal context.
[12]	2023	Twitter, Reddit	Multimodal Transformer	0.931 Twitter, 0.902 Reddit	Struggles with sarcasm and ambiguous emotional content.
[40]	2024	Twitter	Ensemble of NN, XGBoost, and LR	0.905	Inability to capture temporal information.
[41]	2024	Twitter, Reddit	Multimodal Transformer with Time Encoder	0.945 Twitter, 0.913 Reddit	Does not integrate context between text and images.

Table 3.1. Summary of related work on depression detection using multimodal approaches.

3.3 Limitations

Many existing studies on depression detection using deep learning have notable limitations. Some models, like Wang et al. [32], mainly rely on low-level image features, which might not capture deeper emotional signals. Others, such as Li et al. [28], struggle to detect subtle depressive signs that require contextual understanding. Approaches like Yadav et al. [39] overlook the connection between post text and its timing, reducing their effectiveness. Similarly, Bucur et al. [12] face challenges in handling sarcasm and ambiguous emotional expressions, which are common on social media. Anshul et al. [40] also fail to incorporate temporal information, an important aspect of mental health assessment. Most deep learning models focus primarily on text, even though images can provide valuable clues about depression. A more recent multimodal approach by Zafar et al. [41] integrates both text and images but does not establish contextual relationships between them. This thesis seeks to close this gap by incorporating context between text and images to enhance model performance and provide a more comprehensive understanding of depressive cues in social media posts.

CHAPTER 4: DESIGN AND METHODOLOGY

4.1 Overview

This chapter outlines the approach adopted for detecting depression using multimodal data from social media. Our approach uses a deep learning model with a multimodal architecture. It is designed to combine and analyze both text and image information from user posts. The model's structure is inspired by the work of Bucur et al. [12], but we made several improvements to enhance its performance. One major change is replacing the RoBERTa text encoder with CLIP's text encoder, as the MaPLe framework leverages CLIP. Additionally, we introduce learnable context vectors, inspired by the MaPLe framework proposed by Khattak et al. [42], to improve the model's ability to capture contextual information from both textual and visual inputs. By leveraging the complementary information from multimodal data, this methodology seeks to improve the accuracy and robustness of depression detection.

4.2 Image Preprocessing

All input images are adjusted to a fixed size of 224×224 pixels to meet the CLIP-based image encoder's requirements. Normalization is applied using ImageNet mean and standard deviation values. Additionally, all images are converted to RGB format to ensure a uniform input structure. In cases where images are missing, they are replaced with a black 224x224 RGB image. The transformer uses a mask that is set to 1 for valid images and 0 for black images, reflecting the existence of meaningful visual input.

4.3 Model Architecture

Figure 4.1 illustrates the overall model architecture. The subsequent sections provide a detailed explanation of each component.

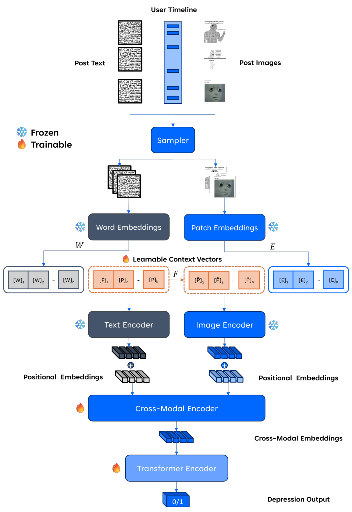

Figure 4.1: Overall architecture of the proposed multimodal depression prediction model.

4.3.1 Sampler

The sampler is designed to extract a random subset of posts from a user's timeline. Given the large volume of data per user, processing every post is impractical and may introduce noise. Inspired by [12], we define a post-sequence P_i as a collection of K sampled posts from the user timeline U_i. Each post comprises text (T), an image (I), and its posting time (Δ):

$$P_i = \Big\{(T^j, I^j, \Delta^j) \sim U_i \mid j \in (1, \ldots, K)\Big\}$$

Here, T^j denotes the text of the j-th post, I^j represents the corresponding image, and Δ^j is the posting time. If a user has fewer than K posts, the sequence is padded to ensure a consistent length.

4.3.2 Text Encoder

The text encoder leverages CLIP's architecture to generate feature representations for textual data. The process is as follows:

1. **Tokenization and Embedding:** The input text is first tokenized, and each token is mapped to a dense vector. This results in the initial word embeddings:

   $$W_0 = [w_1^0, w_2^0, \ldots, w_N^0] \in \mathbb{R}^{N \times d_l},$$

   where N is the number of tokens and d_l is the embedding dimension for the language modality.

2. **Contextual Encoding:** The embeddings are then processed through a series of K transformer layers. At each layer, the token representations are refined by capturing contextual relationships:

$$W_i = L_i(W_{i-1}), \quad i = 1, 2, \ldots, K,$$

where L_i represents the i-th transformer layer. This hierarchical processing ensures that the final representations encode both local and global contextual information.

3. **Projection to Latent Space:** The embedding corresponding to the last token (which aggregates the overall sentence information) from the final transformer layer is then projected into a common latent space:

$$z = \mathrm{TextProj}(w_N^K) \in \mathbb{R}^{d_{vl}},$$

where z is the final text representation, and d_{vl} is the dimensionality of the shared visual-linguistic embedding space.

4.3.3 Image Encoder

The image encoder is designed to extract visual features from images using a transformer-based approach. Its steps are detailed below:

1. **Patch Extraction and Embedding:** The image I is split into M fixed-size patches. Each patch is then linearly projected into an embedding space, yielding:

$$E_0 \in \mathbb{R}^{M \times d_v},$$

where d_v is the dimension of the visual embeddings.

2. **Incorporation of the Class Token:** A learnable class token c_0 is concatenated with the patch embeddings. This class token is intended to capture global image features and serves as a summary representation.

3. **Transformer Processing:** The combined sequence of the class token and patch embeddings is then fed through K transformer blocks:

$$[c_i, E_i] = V_i([c_{i-1}, E_{i-1}]), \quad i = 1, 2, \ldots, K,$$

where V_i denotes the ith transformer block. This process allows the network to learn both local patch features and their global interactions across the image.

4. **Projection to Latent Space:** The class token output from the last transformer layer is used to produce the final image representation, which is then projected into the common latent space:

$$x = \mathrm{ImageProj}(c_K) \in \mathbb{R}^{d_{vl}},$$

where x is the final image embedding aligned with the text representation for multimodal tasks.

4.3.4 Learnable Context Vectors

To enhance the alignment of textual and visual representations, we introduce learnable context vectors, adopting the formulation presented in MaPLe [42]. The equations and architecture described in this section closely follow their original design. This module adjusts the representations to capture depression-related linguistic and visual patterns.

4.3.4.1 Prompt Initialization and Projection

A set of trainable vectors, denoted as $\mathbf{P}$, is introduced to serve as prompts. These vectors are structured to match the dimensionality of the text encoder's embedding space:

$$\mathbf{P} \in \mathbb{R}^{b \times d_l}$$

In this context, b represents the number of prompt tokens, and d_l refers to the embedding size used by the language encoder. The parameters of $\mathbf{P}$ may be initialized either from a predefined textual input or sampled randomly, such as using a normal distribution: $P_{ij} \sim \mathcal{N}(0, 0.02)$.

These language-based prompts $\mathbf{P}$ are incorporated directly into the layers of the text encoder. To adapt them for the image encoder, a corresponding set of visual prompts $\tilde{\mathbf{P}}$ is derived by transforming $\mathbf{P}$ into the visual domain using the coupling mechanism introduced in Equation (4.1).

4.3.4.2 Deep Language Prompting

In the language branch, MaPLe introduces b learnable tokens $\{\mathbf{P}_i \in \mathbb{R}^{d_l}\}_{i=1}^{b}$, which are concatenated with fixed input tokens $\mathbf{W}_0 = [\mathbf{w}_1, \mathbf{w}_2, \ldots, \mathbf{w}_N]$. The input embeddings are represented as:

$$[\mathbf{P}_1, \mathbf{P}_2, \ldots, \mathbf{P}_b, \mathbf{W}_0]$$

These embeddings are processed through the first J transformer layers L_i as follows:

$$[\_, \mathbf{W}_i] = L_i([\mathbf{P}_{i-1}, \mathbf{W}_{i-1}]), \quad \text{for } i = 1, 2, \ldots, J$$

Here $[\,\cdot, \cdot\,]$ is used to represent the concatenation operation.

After the J-th layer, the subsequent layers process the previous layer prompts:

$$[\mathbf{P}_j, \mathbf{W}_j] = L_j([\mathbf{P}_{j-1}, \mathbf{W}_{j-1}]), \quad \text{for } j = J+1, \ldots, K$$

The text features $\mathbf{f}_{\text{text}}$ are obtained by projecting the last token $\mathbf{w}_N^K$:

$$\mathbf{f}_{\text{text}} = \text{TextProj}(\mathbf{w}_N^K)$$

4.3.4.3 Deep Vision Prompting

In the vision branch, MaPLe introduces b learnable tokens $\{\tilde{\mathbf{P}}_i \in \mathbb{R}^{d_v}\}_{i=1}^{b}$, concatenated with image embeddings. The process through the first J transformer layers V_i is:

$$[\mathbf{c}_i, \mathbf{E}_i, \_] = V_i([\mathbf{c}_{i-1}, \mathbf{E}_{i-1}, \tilde{\mathbf{P}}_{i-1}]), \quad \text{for } i = 1, 2, \ldots, J$$

For layers beyond J, the embeddings are processed as:

$$[\mathbf{c}_j, \mathbf{E}_j, \tilde{\mathbf{P}}_j] = V_j([\mathbf{c}_{j-1}, \mathbf{E}_{j-1}, \tilde{\mathbf{P}}_{j-1}]), \quad \text{for } j = J+1, \ldots, K$$

The image features $\mathbf{f}_{\text{image}}$ are obtained by projecting the class token $\mathbf{c}_K$:

$$\mathbf{f}_{\text{image}} = \text{ImageProj}(\mathbf{c}_K)$$

Sharing prompts across successive transformer stages yields better performance than using independent prompts, as the features are highly correlated throughout the network.

4.3.4.4 Vision-Language Prompt Coupling

MaPLe integrates vision and language prompts through a vision-to-language coupling mechanism. Instead of learning vision and language prompts independently, MaPLe projects language prompts into the vision space via a coupling function $F(\cdot)$:

$$\tilde{\mathbf{P}}_k = F_k(\mathbf{P}_k) \tag{4.1}$$
$$[\mathbf{c}_i, \mathbf{E}_i, \_] = V_i([\mathbf{c}_{i-1}, \mathbf{E}_{i-1}, F_{i-1}(\mathbf{P}_{i-1})]), \quad \text{for } i = 1, 2, \ldots, J$$

$$[\mathbf{c}_j, \mathbf{E}_j, \tilde{\mathbf{P}}_j] = V_j([\mathbf{c}_{j-1}, \mathbf{E}_{j-1}, \tilde{\mathbf{P}}_{j-1}]), \quad \text{for } j = J+1, \ldots, K$$
$$\mathbf{f}_{\text{image}} = \text{ImageProj}(\mathbf{c}_K)$$

This function is implemented as a linear layer mapping d_l-dimensional inputs to d_v-dimensional outputs, creating a shared embedding space between the two modalities. The coupled vision and language prompts are learned simultaneously during training, facilitating gradient propagation across both branches and improving mutual adaptation.

MaPLe uses a branch-aware multi-modal prompting strategy. This strategy optimizes context learning for each modality. It also ensures consistent interaction between the vision and language branches.

4.3.5 Projection and Time2Vec Positional Embedding

The textual and visual features, denoted $\mathbf{f}_{\text{text}}$ and $\mathbf{f}_{\text{image}}$, are first projected into a common embedding space using linear projection layers:

$$\tilde{\mathbf{L}} = \mathbf{f}_{\text{text}}\,\mathbf{W}_{\text{text}}, \quad \tilde{\mathbf{V}} = \mathbf{f}_{\text{image}}\,\mathbf{W}_{\text{image}}$$

where $\mathbf{W}_{\text{text}}$ and $\mathbf{W}_{\text{image}}$ are the respective projection matrices. To incorporate the temporal aspect of the user posts, we utilize *time2vec* [43] positional embeddings, as inspired by Bucur et al. [12]. Since social media posts are often asynchronous and irregularly spaced, *time2vec* provides a more effective way to encode the temporal information compared to standard positional embeddings. For any given timestamp τ, the *time2vec* embedding is represented by a $(k+1)$-dimensional vector defined as follows:

$$\mathbf{t2v}(\tau)[i] = \begin{cases} \omega_i\,\mathbf{g}(\tau) + \phi_i, & \text{if } i = 0, \\ \mathbf{F}\Big(\omega_i\,\mathbf{g}(\tau) + \phi_i\Big), & \text{if } 1 \leq i \leq k, \end{cases}$$

where $\mathbf{t2v}(\tau)[i]$ denotes the ith component of the embedding. Here, $\mathbf{F}$ is a periodic activation function (specifically, $\sin(x)$ in our implementation), and the parameters ω_i and ϕ_i are learned during training. To prevent excessively large values of τ, we apply the transformation

$$\mathbf{g}(\tau) = \frac{1}{\tau + \epsilon}, \quad \text{with } \epsilon = 1.$$

These temporal embeddings are then added to the projected features:

$$\hat{\mathbf{L}}^{(0)} = \tilde{\mathbf{L}} + \mathbf{P}_{\text{time}}, \quad \hat{\mathbf{V}}^{(0)} = \tilde{\mathbf{V}} + \mathbf{P}_{\text{time}} \tag{4.2}$$

where $\mathbf{P}_{\text{time}}$ denotes the time-based positional encodings derived from the timestamp sequence Δ^j.

4.3.6 Cross-Modal Embedding Fusion: LXRTXLayer

To effectively integrate the learned textual and visual features, we employ a cross-modal encoder based on the Learning Cross-Modality Encoder Representations from Transformers (LXMERT) architecture [44]. The LXRTXLayer is a key component of this architecture, which facilitates interaction and information exchange between the two modalities through a combination of cross-attention, self-attention, and feed-forward networks.

Let

$$\mathbf{L} \in \mathbb{R}^{N \times T_L \times d} \quad \text{and} \quad \mathbf{V} \in \mathbb{R}^{N \times T_V \times d}$$

represent the language and visual features, respectively, where T_L and T_V are the sequence lengths of the language and visual features, and d is the shared embedding dimension.

4.3.6.1 Cross-Attention

The cross-attention mechanism allows each modality to attend to relevant information from the other modality. The cross-attention from language to vision is computed as:

$$\mathbf{L}_{\text{cross}} = \text{Attention}\Big(\mathbf{Q}_L,\, \mathbf{K}_V,\, \mathbf{V}_V\Big) = \text{softmax}\left(\frac{\mathbf{Q}_L\, \mathbf{K}_V^\top}{\sqrt{d_k}} + M_V\right) \mathbf{V}_V,$$

where $\mathbf{Q}_L$, $\mathbf{K}_V$, and $\mathbf{V}_V$ are the query, key, and value matrices derived from the language and visual features, respectively. Here, d_k is the dimension of the key vectors, and M_V is the attention mask for the visual features. Similarly, the cross-attention from vision to language is:

$$\mathbf{V}_{\text{cross}} = \text{Attention}\Big(\mathbf{Q}_V,\, \mathbf{K}_L,\, \mathbf{V}_L\Big) = \text{softmax}\left(\frac{\mathbf{Q}_V\, \mathbf{K}_L^\top}{\sqrt{d_k}} + M_L\right) \mathbf{V}_L,$$

where $\mathbf{Q}_V$, $\mathbf{K}_L$, and $\mathbf{V}_L$ are derived from the visual and language features, and M_L is the attention mask for the language features.

4.3.6.2 Self-Attention

Following the cross-attention, each modality undergoes self-attention to further refine the representations by allowing them to attend to different parts of themselves:

$$\mathbf{L}_{\text{self}} = \text{SelfAtt}(\mathbf{L}_{\text{cross}}) = \text{Attention}\Big(\mathbf{L}_{\text{cross}},\, \mathbf{L}_{\text{cross}},\, \mathbf{L}_{\text{cross}}\Big),$$

$$\mathbf{V}_{\text{self}} = \text{SelfAtt}(\mathbf{V}_{\text{cross}}) = \text{Attention}\Big(\mathbf{V}_{\text{cross}},\, \mathbf{V}_{\text{cross}},\, \mathbf{V}_{\text{cross}}\Big).$$

4.3.6.3 Feed-Forward Networks (FFN)

Finally, the feature representations for each modality are further transformed using a feed-forward network that introduces non-linearity:

$$\mathbf{L}_{\text{ffn}} = \text{FFN}_{\text{lang}}(\mathbf{L}_{\text{self}}), \quad \mathbf{V}_{\text{ffn}} = \text{FFN}_{\text{visn}}(\mathbf{V}_{\text{self}}),$$

where the FFN is typically defined as:

$$\text{FFN}(\mathbf{X}) = \text{LayerNorm}\Big(\mathbf{X} + \text{Dropout}\Big(\sigma(\mathbf{X}\mathbf{W}_1 + \mathbf{b}_1)\mathbf{W}_2 + \mathbf{b}_2\Big)\Big).$$

Here, σ denotes an activation function, and $\mathbf{W}_1$, $\mathbf{b}_1$, $\mathbf{W}_2$, and $\mathbf{b}_2$ are learnable parameters. The LXRTXLayer outputs the transformed language and visual features:

$$(\mathbf{L}_{\text{ffn}},\, \mathbf{V}_{\text{ffn}}) = \text{LXRTXLayer}(\mathbf{L}, \mathbf{V})$$

4.3.7 Transformer Encoder

The transformer module takes the fused language and visual features from the LXRTXLayer and produces the final prediction.

4.3.7.1 Layer-wise Fusion via LXRTXLayer

A stack of L_{layers} LXRTXLayer modules is applied to iteratively fuse the language and visual features, allowing for deeper interaction between the modalities:

$$\big(\hat{\mathbf{L}}^{(l)},\, \hat{\mathbf{V}}^{(l)}\big) = \text{LXRTXLayer}^{(l)}\Big(\hat{\mathbf{L}}^{(l-1)},\, \hat{\mathbf{V}}^{(l-1)}\Big), \quad l = 1, \ldots, L_{\text{layers}},$$

beginning with the time-encoded inputs $\hat{\mathbf{L}}^{(0)}$ and $\hat{\mathbf{V}}^{(0)}$ as defined in (4.2).

4.3.7.2 Final Encoding and Classification

After passing through the stack of LXRTXLayer modules, the final language features are taken as the cross-modal representation:

$$\mathbf{Z} = \hat{\mathbf{L}}^{(L_{\text{layers}})}.$$

This tensor is then processed by a final transformer encoder consisting of multiple self-attention layers:

$$\mathbf{Z}' = \text{TransformerEncoder}(\mathbf{Z}).$$

To obtain a single representation for the entire sequence of posts, we apply mean pooling over the token dimension (where T denotes the number of tokens in the sequence):

$$\mathbf{z}_{\text{final}} = \frac{1}{T}\sum_{t=1}^{T} \mathbf{Z}'_t.$$

Finally, a linear layer with weight matrix $\mathbf{W}_{\text{class}}$ is used to produce the logits for the classification task:

$$\text{logits} = \mathbf{z}_{\text{final}}\mathbf{W}_{\text{class}}.$$

These logits are then converted into probabilities using a sigmoid activation function, as depression detection is typically framed as a binary classification problem (depressed vs. not depressed):

$$\text{probas} = \sigma\big(\text{logits}\big).$$

A probability greater than or equal to a threshold of 0.5 indicates a prediction of depression.

4.4 Summary of the Pipeline

To summarize, the overall pipeline proceeds as follows:

1. **Textual Processing:** Raw text from sampled user posts is tokenized and embedded using CLIP's text encoder. Learnable context vectors, inspired by MaPLe, are incorporated to enhance the contextual representation. Positional embeddings are added to capture token order.

2. **Visual Processing:** Images from the sampled posts are resized, normalized, and processed using CLIP's image encoder. Visual context tokens are added to enrich the visual representation, and positional embeddings are used to encode patch locations.

3. **Temporal Encoding:** The text and image features are augmented with time2vec positional embeddings to capture the temporal context of the user's posts.

4. **Multimodal Fusion:** The time-encoded textual and visual features are fused using a stack of LXRTXLayer modules, which employ cross-attention and self-attention mechanisms to enable interaction and information exchange between the modalities.

5. **Final Prediction:** The resulting multimodal representation is fed into a transformer encoder, after which mean pooling is applied to extract a single feature vector. This vector is then input to a linear classifier with a sigmoid activation to generate the final depression probability.

CHAPTER 5: EXPERIMENT SETUP DETAILS

This chapter outlines the experimental framework used to detect depression in social media posts through deep learning. It covers the dataset used for training and evaluation, the GPU-based training environment, model training configurations, and the evaluation metrics utilized to assess performance. The objective of this section is to offer a comprehensive explanation of the experimental framework, ensuring transparency and reproducibility.

5.1 Dataset Description

For this study, we utilized the publicly available Twitter dataset provided by Gui et al. [17], which consists of social media posts categorized into two classes: depressed and non-depressed. The multimodal Twitter dataset includes 1,402 depressed users and 1,402 non-depressed users, with posts comprising both text-only content and content that includes both text and images. The detailed distribution of the dataset is as follows:

Class	Text-only	Text+Images
Depressed	213K	19.3K
Non-Depressed	828K	50.6K

Table 5.1. Dataset distribution across classes and modalities.

Each data sample contains a social media post, where textual data consists of raw tweets, and multimodal samples include images attached to the tweets. For image-based samples, we applied resizing and normalization to make the data suitable for processing by the model.

5.2 GPU-Based Training and Inference

To facilitate efficient deep learning model training and inference, we utilized a high-performance computational setup. The training was conducted on an NVIDIA A100

GPU with 40GB of memory, accompanied by a 128GB RAM system powered by an Intel® Core™ i9-13900KF processor. This hardware configuration allowed for optimized batch processing and accelerated computations, significantly reducing training time.

5.3 Implementation Details

The input window size for user posts, denoted by K, was set to 128. The learnable context vector depth was configured with $J = 9$, while the language and vision context vector lengths were both set to 2. The ViT-B/16 CLIP model was employed with parameters $d_l = 512$, $d_v = 768$, and $d_{vl} = 512$. The cross-modal encoder consisted of four cross-encoder layers, each with eight attention heads and an embedding size of 128. Additionally, the self-attention transformer encoder was designed with two layers, each containing eight attention heads and maintaining the same embedding size of 128.

5.4 Hyperparameters

Due to VRAM constraints, we used a batch size of 2 during training. We employed the Adam optimizer with a learning rate of 1×10^{-5}, along with a cyclic learning rate schedule to adaptively adjust the learning rate throughout training. Additionally, we utilized 5-fold cross-validation and trained the model for 50 epochs to ensure robustness and generalization. These hyperparameters were chosen to balance convergence speed and model performance while mitigating overfitting risks.

5.5 Evaluation Metrics

To thoroughly evaluate the effectiveness of our depression detection model, we used a range of metrics that capture different aspects of classification performance. These include:

5.5.1 Accuracy

Accuracy measures the proportion of correctly classified samples relative to the total dataset size and is computed as follows:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \tag{5.1}$$

where TP, TN, FP, and FN denote true positives, true negatives, false positives, and false negatives, respectively.

5.5.2 Precision

Precision, also referred to as Positive Predictive Value (PPV), calculates the ratio of correctly identified positive cases to the total predicted positives:

$$Precision = \frac{TP}{TP + FP} \tag{5.2}$$

5.5.3 Recall

Recall, also known as Sensitivity, measures the proportion of correctly identified positive samples out of all actual positives:

$$Recall = \frac{TP}{TP + FN} \tag{5.3}$$

5.5.4 F1-Score

F1-score provides a balanced measure between precision and recall and is defined as the harmonic mean of both metrics:

$$F1 - Score = 2 \times \frac{Precision \times Recall}{Precision + Recall} \tag{5.4}$$

5.5.5 Area Under the Receiver Operating Characteristic Curve (AUC-ROC)

The AUC-ROC evaluates how well a classifier separates two categories by computing the area beneath the ROC curve. A score close to 1 reflects excellent performance, whereas a score near 0.5 suggests the classifier is performing at the level of random chance.

Applying these metrics allows us to thoroughly evaluate the model's performance in recognizing signs of depression on social media, ensuring the system is both robust and reliable.

5.5.6 Confusion Matrix

A confusion matrix summarizes the classification outcomes by showing the number of true positives, true negatives, false positives, and false negatives. Its structure is illustrated below:

This matrix helps analyze misclassifications, offering insights into the model's strengths and weaknesses by identifying error patterns.

	Predicted Positive	Predicted Negative
Actual Positive	TP	FN
Actual Negative	FP	TN

Table 5.2. Confusion Matrix Representation

CHAPTER 6: RESULTS AND ANALYSIS

6.1 Cross Validation

To assess the effectiveness of our multimodal approach in identifying depression from social media content, we employed a **5-fold cross-validation** strategy. This approach ensures robustness by assessing the model on different training and validation splits, reducing the risk of overfitting. The key performance metrics used for evaluation include **Accuracy**, **Area Under the Curve (AUC)**, **Precision**, **Recall**, and **F1-score**. Additionally, we calculated the **Confusion Matrix** and generated **AUC-ROC curves** for each fold and their average for a more detailed and thorough examination of model performance. The results across the five folds are summarized in Table 6.1.

Fold	**Accuracy**	**AUC**	**Precision**	**Recall**	**F1-score**
Fold-1	0.9696	0.9943	0.9781	0.9607	0.9693
Fold-2	0.9642	0.9902	0.9814	0.9464	0.9636
Fold-3	0.9482	0.9900	0.9771	0.9178	0.9465
Fold-4	0.9589	0.9902	0.9672	0.9500	0.9585
Fold-5	0.9714	0.9961	0.9782	0.9642	0.9642
Average	0.9625	0.9922	0.9765	0.9479	0.9619

Table 6.1. Performance Metrics Across 5-Fold Cross-Validation

The results show that the proposed model consistently achieved high accuracy, with scores ranging from **0.9482** to **0.9714** across different folds. The AUC was notably strong, exceeding **0.99** in every fold, which means the model can reliably tell apart depressive from non-depressive posts. Furthermore, the precision, recall, and F1-score remained consistent, indicating that the model maintains a good trade-off between false positives and false negatives.

6.2 AUC-ROC Curves and Confusion Matrices

To better assess the model's performance, we generated AUC-ROC curves for each fold. These curves illustrate the model's ability to separate the two classes across different decision thresholds. Figure 6.1 displays these curves along with the average AUC-ROC curve. The high AUC values (over 0.99 in all folds) indicate that the model is very effective at separating depressive from non-depressive posts.

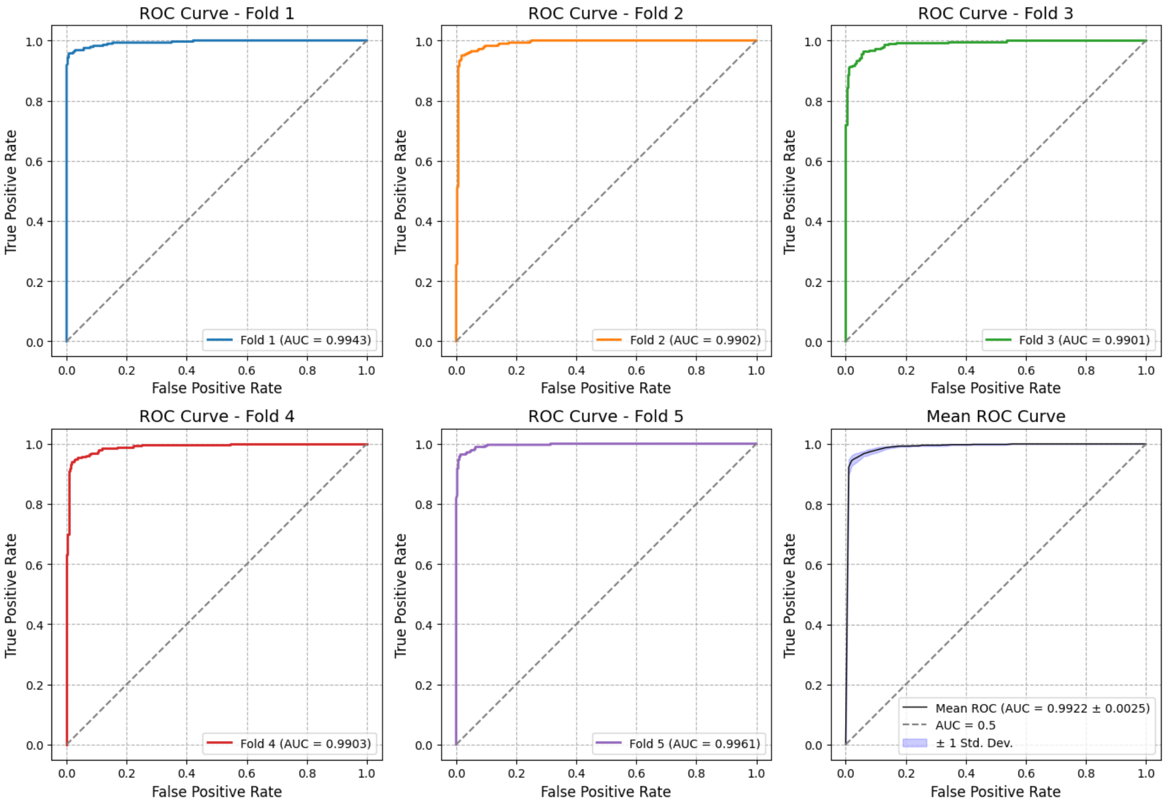

Figure 6.1: AUC-ROC curves for each fold and the mean AUC-ROC curve

We also generated the **Confusion Matrix** for each fold, which breaks down the classification results into true and false positives and negatives, offering a clearer view of model behavior. Figure 6.2 shows the confusion matrices for each fold and the average confusion matrix across the 5 folds, summarizing the model's performance across all folds.

The confusion matrices show that the model accurately identifies depressive posts with few errors.

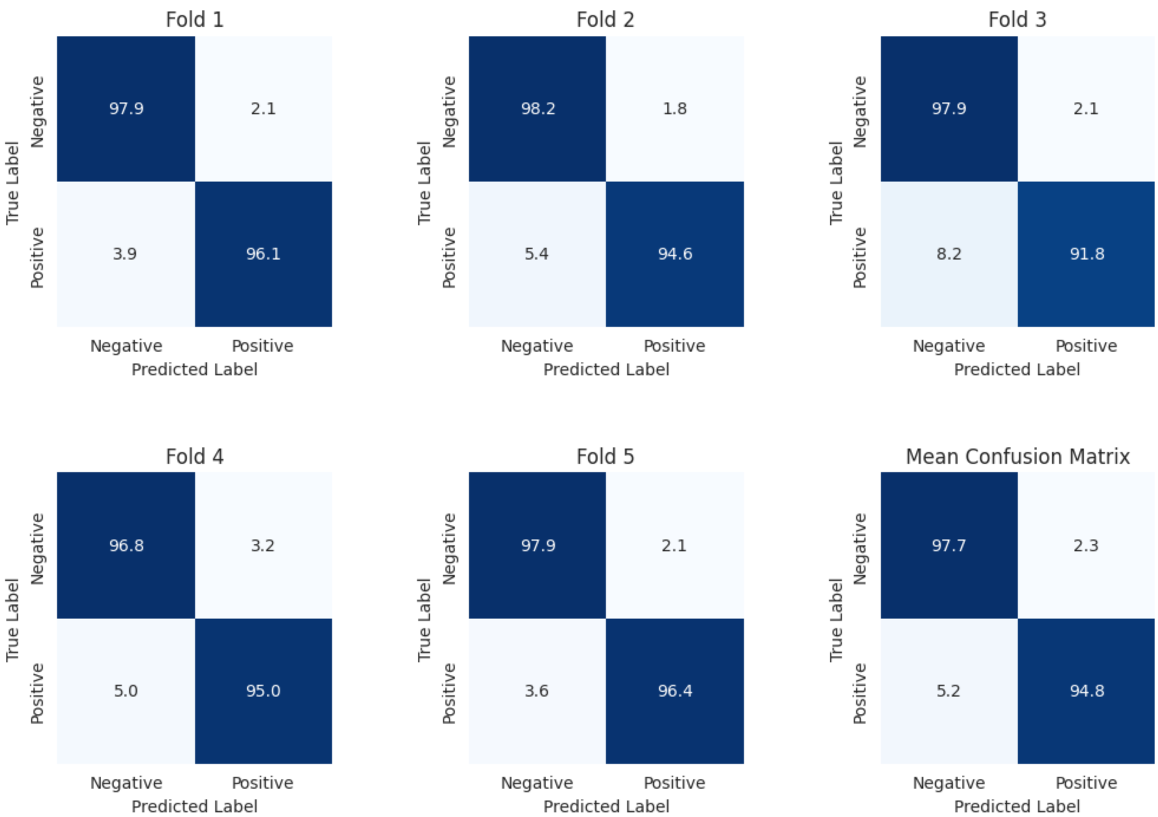

Figure 6.2: Normalized confusion matrices for each of the five folds, along with the average confusion matrix.

6.3 Performance Interpretation

The model's high accuracy, AUC, and confusion matrix results suggest that it works well across different data subsets. High precision shows that the model minimizes false positives, which is important in depression detection to avoid misclassifying healthy individuals. High recall indicates that the model correctly identifies depressive posts with few misses. The F1-score remains above 0.946 in all cases, further supporting the model's reliability.

6.4 Comparison with Existing Literature

In comparison with existing literature, our model achieves superior results. Table 6.2 provides a comparison of our results with the best and second-best reported models in the literature on the Twitter dataset by Gui et al. [17].

Model	Accuracy	AUC	Precision	Recall	F1-score
METN [41]	0.9450	0.9450	0.9450	0.9450	0.9450
Time2Vec Transformer [12]	0.9310	0.9310	0.9310	0.9310	0.9310
Proposed Model (context vectors frozen)	0.9143	0.9574	0.9283	0.8986	0.9131
Proposed Model	**0.9625**	**0.9922**	**0.9765**	**0.9479**	**0.9619**

Table 6.2. Comparison with Existing Literature

Our model outperforms the best results in the literature on all metrics. It improves accuracy by nearly 2% and AUC by over 4%. This indicates that our model better distinguishes depressive content.

We also tested a version of our model with frozen context vectors. In this variant, the context vectors were not updated during training. Although it still beats the Time2Vec Transformer [12] in AUC, its performance drops compared to our full model. Accuracy fell from 0.9625 to 0.9143, and AUC dropped from 0.9922 to 0.9574. This shows that learning task-specific contextual representations is crucial for optimal performance.

A key factor in our model's success is the use of learnable context vectors in both the text and image encoders. These vectors help capture subtle semantic relationships, leading to better feature representation and improved classification. When the context vectors are frozen, the model cannot adapt well to the data, resulting in lower performance metrics. This highlights the importance of dynamically learning contextual representations over using static embeddings.

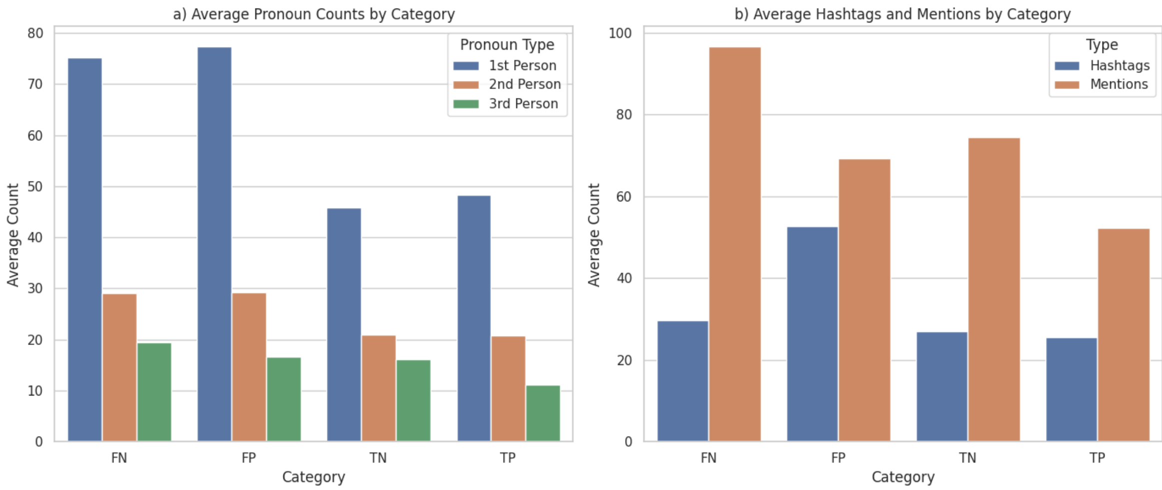

Figure 6.3: Error Analysis of Misclassifications. (a) Average count of first-, second- and third-person pronouns in correctly and incorrectly classified users. (b) Average count of hashtags and mentions in correctly and incorrectly classified users.

6.5 Error Analysis

To better understand the nature of the model's errors, we carried out a qualitative analysis of its misclassifications, examining linguistic features and social media engagement patterns across prediction categories. Our results, obtained using a window size of $K = 128$ and representing the average across a 5-fold cross-validation, show clear differences in the use of personal pronouns, hashtags, and mentions between correctly and incorrectly classified posts.

As shown in Figure 6.3(a), false negatives (FN) and false positives (FP) exhibited significantly higher usage of first-person (75.28 and 77.42, respectively) and second-person pronouns (29.07 and 29.28, respectively) compared to true negatives (TN) and true positives (TP). This finding implies that misclassified users, especially those wrongly identified as non-depressed, often use more self-referential language, which might confuse the model.

In addition, Figure 6.3(b) demonstrates that FN and FP cases have higher average counts of hashtags (29.69 and 52.78, respectively) and mentions (96.67 and 69.25, respectively) compared to correctly classified users. This suggests that increased social media activity may lead to more misclassifications, possibly because the model depends heavily on these engagement features.

6.6 Implications and Practical Applications

The high performance of this approach shows that it is effective in detecting depression. It can be a useful tool for analyzing mental health trends in social media posts. This model may contribute to mental health monitoring, early intervention, and public health research. By automating depression detection reliably, mental health professionals can track trends and offer timely help to those showing signs of depression online.

Moreover, as more people use social media for self-expression, AI-based depression detection systems offer scalable and non-intrusive ways to monitor mental health. Future research could explore integrating this model into real-time monitoring apps, social media platforms, and mental health chatbots to improve early diagnosis and intervention.

CHAPTER 7: CONCLUSION AND FUTURE WORK

7.1 Conclusion

There is growing interest in using social media for detecting signs of depression, given its promise for timely intervention and improved mental health care. While most existing approaches focus primarily on text-based analysis, this research emphasizes the importance of multimodal deep learning by integrating both textual and visual cues.

Our proposed framework introduces learnable context vectors to strengthen the relationship between text and images, allowing for a more comprehensive and context-aware understanding of depression-related content. Our results show that integrating these vectors into both text and image encoders significantly enhances the model's accuracy in detecting depression-related posts, outperforming existing methods.

Beyond presenting a novel framework, this study adds value by presenting an in-depth review of existing research in the field. The review categorizes key datasets, traces the evolution of methodologies, and identifies current challenges. Additionally, we propose a generalized framework for depression detection in social media, offering insights for future research and real-world applications.

7.2 Future Work

Despite the promising findings, several avenues for improvement and further research remain:

- **Exploring Text Preprocessing Techniques:** While no text preprocessing was applied in this study, ensuring a fair comparison with the baselines, future work could systematically evaluate various text preprocessing methods (e.g., token normalization, stop-word removal, or stemming) to determine their impact on model performance.

- **Reducing Computational Load:** The current model requires a significant amount of VRAM and takes considerable time to train. Future research may investigate strategies like pruning, quantization, or knowledge distillation to make the model more efficient while preserving accuracy.
- **Experimenting with Different Text Encoders:** This research relies solely on CLIP for text encoding. Future studies could investigate the effectiveness of alternative encoders like RoBERTa and EmoBERTa, which may provide better representations for sentiment and affective analysis.
- **Expanding to Additional Datasets:** The current study is limited to a Twitter dataset. Future work could incorporate data from Reddit and explore the benefits of using a combined dataset for improved generalizability.
- **Improving Interpretability and Explainability:** While deep learning models achieve high accuracy, their black-box nature remains a challenge. Future work could focus on enhancing model interpretability using techniques like LIME, SHAP, or attention visualization.
- **Real-World Deployment and Clinical Validation:** Collaborating with mental health professionals to validate model predictions and examine whether this model can be effectively integrated into real-world systems for proactive mental health care.

By advancing these areas, future research can further refine multimodal depression detection and contribute to the broader goal of leveraging AI for mental health support ethically and effectively.

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